FIZ 101 · Physics I: Mechanics · Fall 2026

Midterm problem pool

Topics 1–6 · Mehmet Özkan, Istanbul Technical University

The midterm on Tuesday 24 November, during the usual class time, has 10 multiple-choice questions, all taken from this pool of 40 problems. On the exam a question may have different numbers from the ones here, and its choices may be in a different order, so learn how to solve each problem instead of memorizing its answer. The answers are at the end.

No calculators are allowed, and none are needed: the numbers are chosen so that the arithmetic can be done by hand. Take $g = 10\,\mathrm{m/s^2}$ throughout, and ignore air resistance unless a problem says otherwise. Each problem is based on the lecture notes of its topic.

Topic 1 · Units, physical quantities and vectors

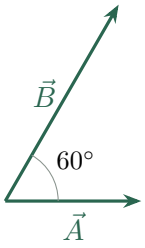
1. An object is released from rest at a height h above the ground and falls freely. Suppose that the speed v with which it reaches the ground depends only on h, on the mass m of the object and on the magnitude g of the acceleration due to gravity, so that $v \propto h^{\alpha} m^{\beta} g^{\gamma}$. Which expression for v does dimensional analysis give? (C is a dimensionless constant.)

(A) $C\sqrt{h/g}$ (B) $C\,gh$ (C) $C\sqrt{gh}$ (D) $C\sqrt{mgh}$ (E) $C\sqrt{g/h}$

2. Which of the following measured lengths has exactly three significant figures?

(A) $0.052\,\mathrm{m}$ (B) $12.05\,\mathrm{m}$ (C) $5.2 \times 10^2\,\mathrm{m}$ (D) $0.5020\,\mathrm{m}$ (E) $0.0520\,\mathrm{m}$

3. Two vectors $\vec{A}$ and $\vec{B}$, of magnitudes $A = 3\,\mathrm{m}$ and $B = 5\,\mathrm{m}$, are drawn from the same point with an angle of 60° between them, as in the figure. Use $\cos 60^\circ = \frac{1}{2}$ and $\sin 60^\circ = \frac{\sqrt{3}}{2}$. What is the magnitude of their sum $\vec{A} + \vec{B}$?



(A) $2\,\mathrm{m}$ (B) $\sqrt{19}\,\mathrm{m}$ (C) $\sqrt{34}\,\mathrm{m}$ (D) $7\,\mathrm{m}$ (E) $8\,\mathrm{m}$

4. The vector $\vec{A} = -8\,\hat{\imath} + 6\,\hat{\jmath}$ lies in the xy plane. What angle θ does it make with the positive x axis, measured from that axis towards the positive y axis, with $0 \le \theta < 360^\circ$? Use $\sin 37^\circ = \cos 53^\circ = 0.6$ and $\cos 37^\circ = \sin 53^\circ = 0.8$.

(A) 37° (B) 53° (C) 127° (D) 143° (E) 323°

5. For what value of the constant b are the vectors $\vec{A} = 2\,\hat{\imath} + b\,\hat{\jmath} + 3\,\hat{k}$ and $\vec{B} = 4\,\hat{\imath} - 2\,\hat{\jmath} + 2\,\hat{k}$ perpendicular to each other?

(A) -7 (B) 3 (C) 4 (D) 7 (E) 14

6. What is the vector product $\vec{A} \times \vec{B}$ of the vectors $\vec{A} = 2\,\hat{\imath} + 3\,\hat{\jmath}$ and $\vec{B} = 4\,\hat{k}$?

(A) $\vec{0}$ (B) $12\,\hat{\imath} - 8\,\hat{\jmath}$ (C) $12\,\hat{\imath} + 8\,\hat{\jmath}$ (D) $-12\,\hat{\imath} + 8\,\hat{\jmath}$ (E) $8\,\hat{\imath} + 12\,\hat{\jmath}$

Topic 2 · Kinematics: motion in one, two and three dimensions

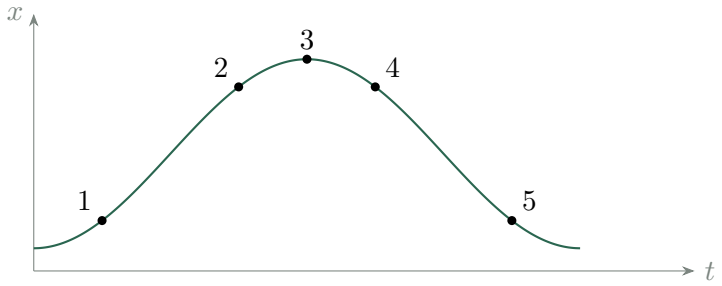
7. In an xy coordinate system a vector is $\vec{A} = (7\,\text{m})\,\hat{\imath} + (24\,\text{m})\,\hat{\jmath}$. A second coordinate system, $x'y'$, has the same origin, but its axes are rotated in the xy plane with respect to the x and y axes (the vector lies in this plane, so it has no z component in either system). In the second system the x' component of the vector is $A'_x = 15\,\text{m}$. What is the magnitude of its y' component, $|A'_y|$?

(A) 10 m (B) 16 m (C) 20 m (D) 24 m (E) 25 m

8. A cyclist rides along a straight road. She rides east at a constant 5 m/s for 60 s, then turns around and rides west at a constant 3 m/s for 20 s. Take east as the positive x direction. What is the x component of her average velocity, $v_{\text{av-}x}$, for the whole trip?

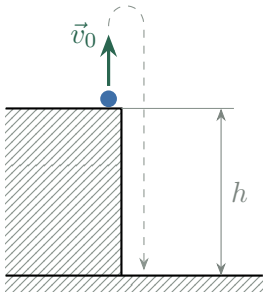
(A) −3 m/s (B) 1 m/s (C) 3 m/s (D) 4 m/s (E) 4.5 m/s

9. The figure shows the position–time (x–t) graph of an object moving along the x axis. At which of the marked points is the object moving in the positive x direction and slowing down?



(A) At point 1 (B) At point 2 (C) At point 3 (D) At point 4 (E) At point 5

10. A stone is thrown straight up with speed $v_0 = 20\,\text{m/s}$ from the edge of a cliff of height $h = 25\,\text{m}$. On its way down it misses the edge and falls to the ground at the foot of the cliff. How long after it is thrown does it hit the ground? Ignore air resistance.



(A) 1 s (B) $\sqrt{5}$ s (C) 4 s (D) 5 s (E) $(4 + \sqrt{5})$ s

11. A particle moves in space with position vector $\vec{r}(t) = \alpha t^3\,\hat{\imath} + \beta t\,\hat{\jmath} + \gamma\,\hat{k}$, where α, β and γ are constants. What is its acceleration $\vec{a}(t)$?

(A) $3\alpha t^2\,\hat{\imath} + \beta\,\hat{\jmath}$ (B) $6\alpha t\,\hat{\imath} + \beta\,\hat{\jmath}$ (C) $3\alpha t\,\hat{\imath}$ (D) $6\alpha\,\hat{\imath}$ (E) $6\alpha t\,\hat{\imath}$

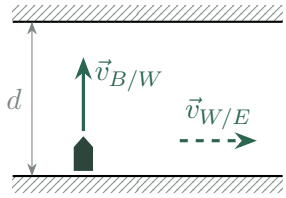
12. A ball is kicked from level ground with speed $v_0 = 25\,\text{m/s}$ and lands back on the ground $t_{\text{tot}} = 4\,\text{s}$ later. How far from the kicking point does it land? Ignore air resistance.

(A) 20 m (B) 30 m (C) 60 m (D) 80 m (E) 100 m

13. A small ball on a string moves at constant speed around a horizontal circle of radius r, making f revolutions per second. What is the magnitude of its acceleration?

(A) $4\pi^2 f^2 r$ (B) $2\pi f r$ (C) 0 (D) $4\pi^2 r/f^2$ (E) $f^2 r$

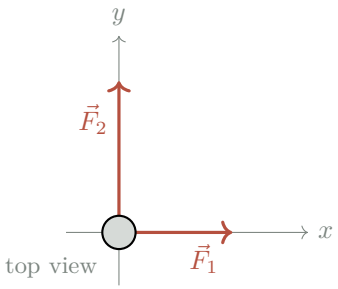
14. A motorboat crosses a river of width $d = 48\,\text{m}$. It heads straight across, perpendicular to the banks, at $v_{B/W} = 8\,\text{m/s}$ relative to the water, and the water flows parallel to the banks at $v_{W/E} = 6\,\text{m/s}$ relative to the Earth (subscripts: B boat, W water, E Earth). What is the straight-line distance from the boat's starting point to the point where it reaches the other bank?



(A) 36 m (B) 48 m (C) 60 m (D) 80 m (E) 84 m

Topic 3 · Newton's laws of motion

15. A puck of mass $m = 2\,\text{kg}$ slides on a horizontal frictionless table, and its weight is balanced by the normal force of the table. Seen from above, two horizontal forces act on it, $\vec{F}_1 = (6\,\text{N})\,\hat{\imath}$ and $\vec{F}_2 = (8\,\text{N})\,\hat{\jmath}$, where the x and y axes lie in the plane of the table. What is the magnitude of the acceleration of the puck?



(A) $1\,\text{m/s}^2$ (B) $5\,\text{m/s}^2$ (C) $7\,\text{m/s}^2$ (D) $10\,\text{m/s}^2$ (E) $50\,\text{m/s}^2$

16. A ship moves in a straight line at a constant speed $v = 12\,\text{m/s}$ relative to the shore. A sailor on its deck throws a ball straight up, as seen on the ship, with a speed $u = 10\,\text{m/s}$ relative to the ship. Ignore air resistance. As seen from the shore, how far does the ball move horizontally between leaving the sailor's hand and coming back down to the height of the hand?

(A) 0 m (B) 12 m (C) 20 m (D) 24 m (E) 44 m

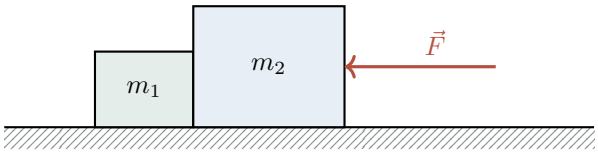
17. An astronaut weighs 600 N on the Earth. On the Moon the acceleration due to gravity is one sixth of its value on the Earth. What net horizontal force gives the astronaut a horizontal acceleration of $2\,\text{m/s}^2$ on the Moon?

(A) 20 N (B) 100 N (C) 120 N (D) 720 N (E) 1200 N

18. A crate rests on a horizontal frictionless surface. At time zero a rope starts to pull it horizontally with a constant force of magnitude $F = 30\,\text{N}$. In the first $t = 3\,\text{s}$ the crate moves a distance $d = 9\,\text{m}$. What is the mass of the crate?

(A) 3 kg (B) 5 kg (C) 10 kg (D) 15 kg (E) 30 kg

19. Two blocks, of masses m_1 and m_2, stand side by side and touch each other on a horizontal frictionless surface. A horizontal force of magnitude F pushes block 2 to the left, and both blocks move together. What is the magnitude of the force that block 1 exerts on block 2?



(A) $\dfrac{F}{2}$ (B) $\dfrac{m_1 F}{m_2}$ (C) $\dfrac{m_2 F}{m_1 + m_2}$ (D) $\dfrac{m_1 F}{m_1 + m_2}$ (E) F

20. A book lies at rest on a horizontal table, and the table stands on the floor. Which of the following pairs of forces is an action–reaction pair of Newton's third law?

(A) The weight of the book and the upward push of the table on the book.
(B) The upward push of the table on the book and the downward push of the book on the table.
(C) The weight of the book and the downward push of the book on the table.
(D) The downward push of the book on the table and the upward push of the floor on the table.
(E) The downward push of the book on the table and the upward pull of the book on the Earth.

Topic 4 · Applying Newton's laws

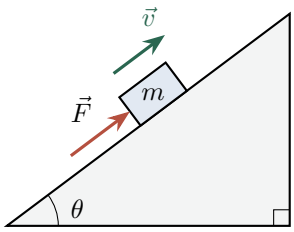
21. A student of mass $m = 60\,\text{kg}$ stands on a bathroom scale in an elevator. The elevator is moving downward at $v = 4\,\text{m/s}$ when it starts to slow down; it slows down at a constant rate and stops after moving a further $d = 4\,\text{m}$. What does the scale read (the magnitude of the force it exerts on the student) while the elevator slows down?

(A) 120 N (B) 480 N (C) 600 N (D) 720 N (E) 840 N

22. A box of mass $m = 20\,\text{kg}$ rests on a horizontal floor. The coefficients of friction between the box and the floor are $\mu_s = 0.5$ and $\mu_k = 0.3$. You push the box with a horizontal force of magnitude $F = 80\,\text{N}$. What is the magnitude of the friction force that the floor exerts on the box while you push?

(A) 0 N (B) 40 N (C) 60 N (D) 80 N (E) 100 N

23. You push a block of mass $m = 5\,\text{kg}$ up a rough incline of angle $\theta = 37°$ with a force $\vec{F}$ parallel to the incline, so that the block moves up the incline with constant velocity. The coefficient of kinetic friction between the block and the incline is $\mu_k = 0.4$. Use $\sin 37° = 0.6$ and $\cos 37° = 0.8$. What is the magnitude of $\vec{F}$?



(A) 14 N (B) 30 N (C) 46 N (D) 50 N (E) 52 N

24. A crate sits on the flat, horizontal bed of a truck. The truck drives forward on a straight, level road and speeds up, and the crate does not slide on the bed. Which statement about the friction force that the bed exerts on the crate is true?

(A) There is no friction force, because the crate does not slide on the bed.
(B) It is kinetic friction, and it points backward.
(C) It is static friction, and it points backward, opposite to the motion.
(D) It is static friction, and it points forward.
(E) It is kinetic friction, and it points forward.

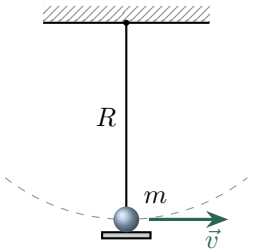
25. Two balls of the same size and shape are dropped from rest from a great height. One ball has four times the mass of the other. For both balls the air resistance has magnitude $f = Dv^2$, with the same constant D. What is the terminal speed of the heavier ball divided by the terminal speed of the lighter ball?

(A) $\frac{1}{2}$ (B) 1 (C) 2 (D) 4 (E) 16

26. A car takes a flat (unbanked) curve of radius $R = 50\,\text{m}$ at a constant speed $v = 20\,\text{m/s}$. What is the smallest coefficient of static friction between the tires and the road for which the car can take the curve without sliding?

(A) 0.04 (B) 0.4 (C) 0.8 (D) 1.25 (E) 8

27. A child of mass m sits on a swing whose ropes have length R. Treat the child as a particle that moves on a circle of radius R. At the lowest point of the swing the child's speed is v. What is the magnitude n of the force that the seat exerts on the child at that moment?



(A) mg (B) $mg + \dfrac{mv^2}{R}$ (C) $mg - \dfrac{mv^2}{R}$ (D) $\dfrac{mv^2}{R}$ (E) $\dfrac{mv^2}{R} - mg$

Topic 5 · Work and kinetic energy

28. A rope pulls a box of mass $m = 5\,\text{kg}$, initially at rest on a horizontal floor, with a constant horizontal force of magnitude $F = 30\,\text{N}$, and the box slides along the floor. The coefficient of kinetic friction between the box and the floor is $\mu_\text{k} = 0.4$. What is the speed of the box when it has moved a distance $d = 9\,\text{m}$?

(A) $3\,\text{m/s}$ (B) $3\sqrt{2}\,\text{m/s}$ (C) $6\,\text{m/s}$ (D) $6\sqrt{3}\,\text{m/s}$ (E) $6\sqrt{5}\,\text{m/s}$

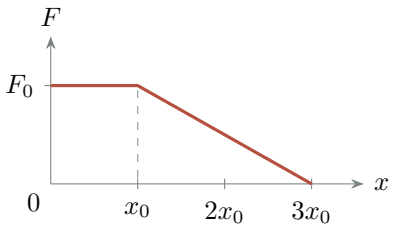
29. A car moves at $v_0 = 20\,\text{m/s}$ on a level road. The driver brakes so hard that the wheels lock, and the car skids to a stop. The coefficient of kinetic friction between the tires and the road is $\mu_\text{k} = 0.5$. Ignore air resistance. How far does the car skid?

(A) 2 m (B) 10 m (C) 20 m (D) 40 m (E) 80 m

30. You lift a backpack of mass m from the floor and put it on a table, a height h above the floor. The backpack is at rest before you lift it and at rest on the table. Ignore air resistance. Which statement is correct?

(A) The total work done on the backpack is zero.
(B) The total work done on the backpack is mgh.
(C) You do no work on the backpack, because it starts and ends at rest.
(D) Gravity does positive work on the backpack.
(E) The net force on the backpack is zero during the whole lift.

31. A block of mass m starts from rest at $x = 0$ on a frictionless horizontal surface. A horizontal force F along the x axis acts on it, and changes with the position x as shown in the graph. Take $F_0 = 8\,\text{N}$, $x_0 = 2\,\text{m}$ and $m = 4\,\text{kg}$. What is the speed of the block at $x = 3x_0$?

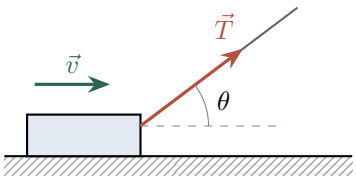


(A) $2\sqrt{2}\,\text{m/s}$ (B) $2\sqrt{3}\,\text{m/s}$ (C) $4\,\text{m/s}$ (D) $2\sqrt{6}\,\text{m/s}$ (E) $16\,\text{m/s}$

32. One end of a horizontal spring with force constant k is fixed. The spring is already stretched by d from its relaxed length. How much work must you do to stretch it slowly further, until it is stretched by $2d$?

(A) $\frac{1}{2}kd^2$ (B) kd^2 (C) $\frac{3}{2}kd^2$ (D) $2kd^2$ (E) $3kd^2$

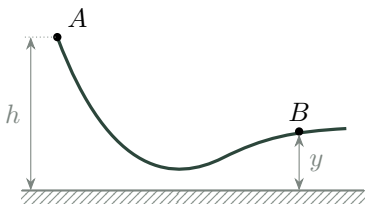
33. A person pulls a sled across level snow at a constant speed $v = 2\,\text{m/s}$. The rope makes an angle $\theta = 37°$ above the horizontal, and the tension in it is $T = 50\,\text{N}$. At what rate does the rope do work on the sled? Use $\sin 37° = 0.6$ and $\cos 37° = 0.8$.



(A) $0\,\text{W}$ (B) $60\,\text{W}$ (C) $80\,\text{W}$ (D) $100\,\text{W}$ (E) $140\,\text{W}$

Topic 6 · Potential energy and energy conservation

34. A small block is released from rest at point A of a frictionless track, a height $h = 5\,\text{m}$ above the ground. What is its speed when it passes point B of the track, a height $y = 1.8\,\text{m}$ above the ground?

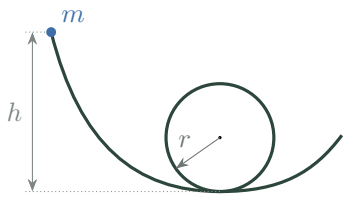


(A) $4\,\text{m/s}$ (B) $6\,\text{m/s}$ (C) $8\,\text{m/s}$ (D) $10\,\text{m/s}$ (E) $64\,\text{m/s}$

35. A ball is dropped from rest from a window and falls to the ground. Ignore air resistance. Ali and Elif both describe the fall with $U_{\text{grav}} = mgy$ and the $+y$ axis upward, but Ali puts $y = 0$ at the ground and Elif puts $y = 0$ at the window. Which of these quantities do they find to be different?

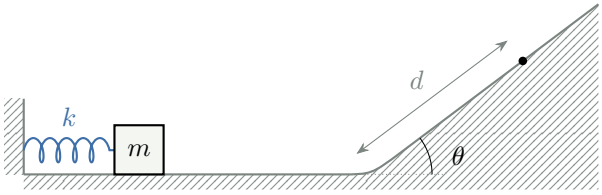
(A) The speed of the ball just before it reaches the ground
(B) The work done by gravity on the ball during the fall
(C) The change in the gravitational potential energy of the ball during the fall
(D) The kinetic energy of the ball just before it reaches the ground
(E) The gravitational potential energy of the ball at the window

36. A small car of mass $m = 0.5\,\text{kg}$ is released from rest on a frictionless track, a height $h = 7\,\text{m}$ above the lowest point of a vertical circular loop of radius $r = 2\,\text{m}$. What is the magnitude of the normal force that the track exerts on the car at the top of the loop?



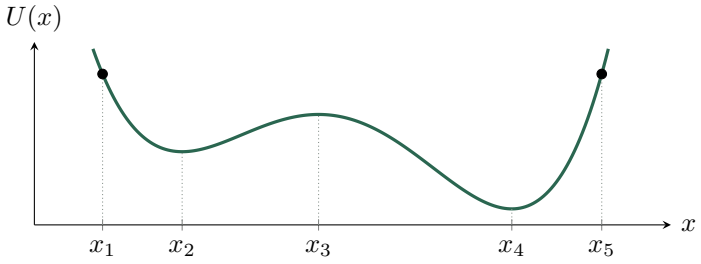
(A) $5\,\text{N}$ (B) $10\,\text{N}$ (C) $15\,\text{N}$ (D) $20\,\text{N}$ (E) $30\,\text{N}$

37. A block of mass $m = 0.5\,\text{kg}$ is pushed against a spring of force constant $k = 150\,\text{N/m}$, compressing it by $x = 0.2\,\text{m}$, and released from rest. It leaves the spring, slides along a frictionless horizontal floor and then up a frictionless ramp that makes the angle $\theta = 37°$ with the horizontal; the floor curves smoothly into the ramp. Use $\sin 37° = 0.6$ and $\cos 37° = 0.8$. What distance d along the ramp, measured from its foot as in the figure, does the block slide before it stops for an instant?



(A) $0.6\,\text{m}$ (B) $0.75\,\text{m}$ (C) $1\,\text{m}$ (D) $2\,\text{m}$ (E) $10\,\text{m}$

38. A particle moves along the x axis, and the only force that does work on it is conservative, with the potential energy $U(x)$ shown. U has the same value at x_1 and x_5. The particle is released from rest at x_1. Which statement is true?

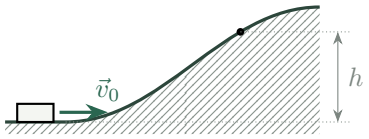


(A) It stops for an instant at x_2, where the force on it is zero.
(B) It cannot get past x_3, because x_3 is the top of a hill of U.
(C) It moves back and forth between x_1 and x_5.
(D) Its speed is greatest at x_3.
(E) The point x_3 is a position of stable equilibrium.

39. A particle moves in a region where the potential energy is $U(x, y, z) = ax^2y + byz$, where a and b are constants. What is the force $\vec{F}$ on the particle at the point (x, y, z)?

(A) $\vec{F} = -2axy\,\hat{\imath} - bz\,\hat{\jmath} - by\,\hat{k}$
(B) $\vec{F} = 2axy\,\hat{\imath} + (ax^2 + bz)\,\hat{\jmath} + by\,\hat{k}$
(C) $\vec{F} = -2axy\,\hat{\imath} - (ax^2 + bz)\,\hat{\jmath}$
(D) $\vec{F} = -2axy\,\hat{\imath} - (ax^2 + bz)\,\hat{\jmath} - by\,\hat{k}$
(E) $\vec{F} = -2axy\,\hat{\imath} - (ax^2 + by)\,\hat{\jmath} - bz\,\hat{k}$

40. A sled of mass $m = 10\,\text{kg}$ moves at $v_0 = 10\,\text{m/s}$ at the bottom of a snowy hill. It slides up the hill and stops at a height $h = 4\,\text{m}$ above the bottom. Ignore air resistance. How much work does the friction force of the snow do on the sled while it slides up?



(A) $-900\,\text{J}$ (B) $-500\,\text{J}$ (C) $-400\,\text{J}$ (D) $-100\,\text{J}$ (E) $+100\,\text{J}$

Answers

1. (C) $v = C\sqrt{gh}$
2. (E) $0.0520\,\mathrm{m}$
3. (D) $7\,\mathrm{m}$
4. (D) $\theta = 143^\circ$
5. (D) $b = 7$
6. (B) $\vec{A} \times \vec{B} = 12\,\hat{\imath} - 8\,\hat{\jmath}$
7. (C) $|A'_y| = 20\,\mathrm{m}$
8. (C) $v_{\text{av-}x} = 3\,\mathrm{m/s}$
9. (B) At point 2
10. (D) $t = 5\,\mathrm{s}$
11. (E) $\vec{a} = 6\alpha t\,\hat{\imath}$
12. (C) $60\,\mathrm{m}$
13. (A) $a_{\text{rad}} = 4\pi^2 f^2 r$
14. (C) $60\,\mathrm{m}$
15. (B) $5\,\mathrm{m/s^2}$
16. (D) $24\,\mathrm{m}$
17. (C) $120\,\mathrm{N}$
18. (D) $15\,\mathrm{kg}$
19. (D) $\dfrac{m_1 F}{m_1 + m_2}$
20. (B) Table on book and book on table
21. (D) $720\,\mathrm{N}$
22. (D) $80\,\mathrm{N}$
23. (C) $46\,\mathrm{N}$
24. (D) Static friction, pointing forward
25. (C) 2
26. (C) $\mu_{\text{s}} = 0.8$
27. (B) $n = mg + mv^2/R$
28. (C) $6\,\mathrm{m/s}$
29. (D) $40\,\mathrm{m}$
30. (A) The total work is zero
31. (C) $4\,\mathrm{m/s}$
32. (C) $\frac{3}{2}kd^2$
33. (C) $80\,\mathrm{W}$
34. (C) $8\,\mathrm{m/s}$
35. (E) The gravitational potential energy at the window
36. (B) $10\,\mathrm{N}$
37. (C) $1\,\mathrm{m}$
38. (C) It moves back and forth between x_1 and x_5
39. (D) $\vec{F} = -2axy\,\hat{\imath} - (ax^2 + bz)\,\hat{\jmath} - by\,\hat{k}$
40. (D) $-100\,\mathrm{J}$